OPERATING SYSTEMS

ORANGE PROBLEM

PES Version Control System (PES-VCS)

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SECTION : K

AIM:

To implement a simplified Git-like version control system supporting object storage, indexing, tree construction, and commit management.

Screenshot 1A – Object Test Output

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Screenshot 1B – Object Storage Directory Structure

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

Screenshot 2A – Tree Test Output

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Screenshot 2B – Raw Object (Hex Dump)

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213
ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

Screenshot 3A – Repository Initialization and Staging

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ ./pes init
echo "hello" > file.txt
./pes add file.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h
```

Screenshot 3B – Index File Contents

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ cat .pes/index
100777 file.txt 1776835377 6 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$
```

Screenshot 4A – Commit Creation

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ ./pes commit -m "first commit"
echo "second" >> file.txt
./pes add file.txt
./pes commit -m "second commit"
echo "third" >> file.txt
./pes add file.txt
./pes commit -m "third commit"
./pes log
Committed: 50bcf38a9672... first commit
Committed: 553cf61fb25e... second commit
Committed: 77b6fbcc6c95... third commit
```

Screenshot 4B – Object Growth

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/39/de25b80c915eee385fdac03ef3e71ad36a9f383557aeb8b71169403dd30082
.pes/objects/50/bcf38a9672c6ba3303f8d8e280af18e37c8800a5e98a0fea38c9852e029827
.pes/objects/53/936109158bccfbbba73de54297ed22203c8741746156a83c2c67f1bce2ce2e
.pes/objects/55/3cf61fb25e14a03062f9d15d60c946b6f62f37ce29ba04b8f2da05d8770f94
.pes/objects/77/b6fbcc6c951c5cda235ad94ed4df966fb0a700ef8e1cc37bfdc2a521a8d101
.pes/objects/b3/b2314e14a79e451328eff94faef74bafe1c22a54033753678ad2488cc559b3
.pes/objects/bf/13f2b562ec02f8fcdaac9e76b5e51ab85af8307f917322862facb799e296a3
.pes/objects/bf/7196aaa94bf03bb4c074e36d169588b72565922508f27b8b2b4288e77c8b5e
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/refs/heads/main
```

Screenshot 4C – HEAD and Branch Reference

```
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
77b6fbcc6c951c5cda235ad94ed4df966fb0a700ef8e1cc37bfdc2a521a8d101
ref: refs/heads/main
dath@LAPTOP-8VTHFTV2:/mnt/c/Users/GURUDATH A N/PES2UG24CS633-pes-vcs$
```

Integration Test Output

```
--- First Commit ---
Committed: 80507ec3ee9d... Initial commit

Log after first commit:
No commits yet.

--- Second Commit ---
Committed: ca7d498aaecd... Update file.txt

--- Third Commit ---
Committed: 66e70b62e354... Add farewell

--- Full History ---
No commits yet.

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
66e70b62e35465c0a122bcaa3a838fa93f24485c49cd5a9d5ff263cc143d44e6

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/66/e70b62e35465c0a122bcaa3a838fa93f24485c49cd5a9d5ff263cc143d44e6
.pes/objects/80/507ec3ee9d8eb2c93a8190a802ac35b7462472782b032d41108c2a5fd73e9f
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ca/7d498aaecd2de9240a8de5751d3ae4f3869bb02e1829d2fe1dade0d7bede79
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Q5.1 – What is branching in version control?

Branching in version control allows multiple lines of development to exist simultaneously.

Each branch is a pointer to a specific commit, enabling independent development without

affecting the main codebase. Developers can create branches to work on new features or fixes

and later merge them back into the main branch.

Q5.2 – How are branches represented internally?

Internally, a branch is represented as a reference (pointer) to a commit.

In this system, branches are stored as files under .pes/refs/heads/, where each file

contains the hash of the latest commit in that branch. The HEAD file points to the

current active branch.

Q5.3 – What is merging?

Merging is the process of combining changes from one branch into another.

It integrates different lines of development into a single branch.

In version control systems like Git, merging ensures that all changes from

different branches are incorporated into the main project history.

Q6.1 – What is garbage collection in version control?

Garbage collection is the process of removing unused or unreachable objects

from the repository. Objects that are not referenced by any commit, tree, or

branch are considered unreachable and can be safely deleted to free storage space.

Q6.2 – Why is garbage collection important?

Garbage collection is important because it helps in maintaining storage efficiency

by removing unnecessary data. It reduces repository size, improves performance,

and ensures that only relevant objects are retained in the system.

The following files are implemented as part of this project:

- object.c – Object storage implementation
- tree.c – Tree construction and serialization
- index.c – Staging area (index) implementation
- commit.c – Commit creation and history tracking
- pes.c – Command-line interface

All source code is available in the GitHub repository.